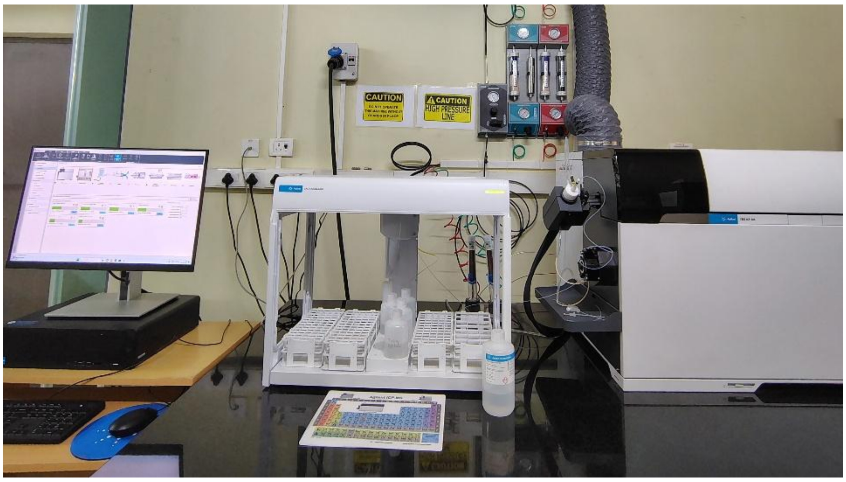
Heavy Metals Analysis @ ICWaR



Our laboratory offers high-precision heavy metal analysis using the Agilent 7850 ICP-MS (Inductively Coupled Plasma Mass Spectrometry). This advanced analytical technique enables detecting and quantifying trace and ultra-trace levels of heavy metals in water, soil, industrial effluents, and environmental samples. Heavy metal contamination poses significant risks to human health and ecosystems, making accurate monitoring essential for compliance with regulatory standards.

The Agilent 7850 ICP-MS is a high-performance instrument designed for superior sensitivity, accuracy, and efficiency, featuring:

- ISIS 3: High-Productivity Option – Enhances sample throughput and reduces analysis time.
- ORS4 Collision/Reaction Cell – Uses helium (He) mode for efficient interference removal, ensuring accurate quantification of elements in complex matrices.
- ODS (Orthogonal Detector System) – Provides a 10-order dynamic range, allowing simultaneous measurement of major and trace elements in a single run.

Key Advantages of ICP-MS Analysis

✓ Simplified Workflow – Pre-set methods and auto-optimization for easy method setup and consistent performance.

✓ Enhanced Matrix Tolerance – HMI technology enables direct analysis of samples with up to 3% dissolved solids, minimising manual or auto-dilution.

✓ Reliable Results in Complex Matrices – The ORS4 cell effectively removes polyatomic interferences for unmatched accuracy.

✓ High Productivity & Faster Turnaround – The ISIS 3 system and automated method optimisation reduce sample processing time.

Heavy metal analysis services are ideal for drinking water quality assessment, industrial effluent monitoring, environmental impact studies, and regulatory compliance testing. Whether you require routine screening or specialised trace metal analysis, this laboratory ensures high-quality, reliable, and cost-effective results.